Verification
01_H_Simulator_FVM_1D_Incompressible_Homogeneous

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical solution for one-dimensional incompressible single-phase flow in a homogeneous porous medium.

2 Governing Equation

For steady-state incompressible flow with no gravity and no source terms, conservation of mass combined with Darcy's law gives

$$\frac{\partial}{\partial x}\left(-\frac{k}{\mu}\frac{\partial P}{\partial x}\right) = 0, \tag{1}$$

where

- P is pressure [Pa]
- k is permeability [m^2]
- μ is fluid viscosity [Pa·s]

The boundary conditions are,

$$P(0) = P_L, \quad P(L_x) = P_R. \tag{2}$$

3 Analytical Solution

Integrating the governing equation twice yields a linear pressure distribution,

$$P(x) = P_L + \frac{P_R - P_L}{L_x}x. \tag{3}$$

The corresponding Darcy flux is

$$q = -\frac{k}{\mu}\frac{P_R - P_L}{L_x}, \tag{4}$$

which is constant throughout the domain.

4 Simulation Parameters

The simulation parameters are given in Table 1.

Parameter	Symbol	Value
Number of cells	N_x	100
Domain length	L_x	1 m
Permeability	k	1 m^2
Viscosity	μ	1 Pa·s
Left pressure	P_L	10×10^6 Pa
Right pressure	P_R	0 Pa

Table 1: Simulation parameters.

5 Results

5.1 Pressure Distribution

The pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure 2, match well.

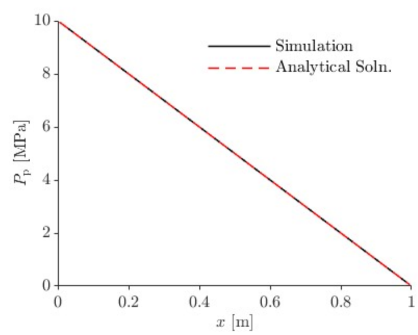

Figure 1: Comparison between analytical and numerical pressure solutions for the MATLAB simulator.

5.2 Darcy Flux

The Darcy flux solutions for both the MATLAB simulator, Figure 3, and the Python simulator, Figure 4, match well.

6 Error Analysis

The maximum pressure relative error is defined as

$$E_P = \max_i \left| \frac{P_{\text{num},i} - P_{\text{ana},i}}{q_{\text{ana},i}} \right| . \tag{5}$$

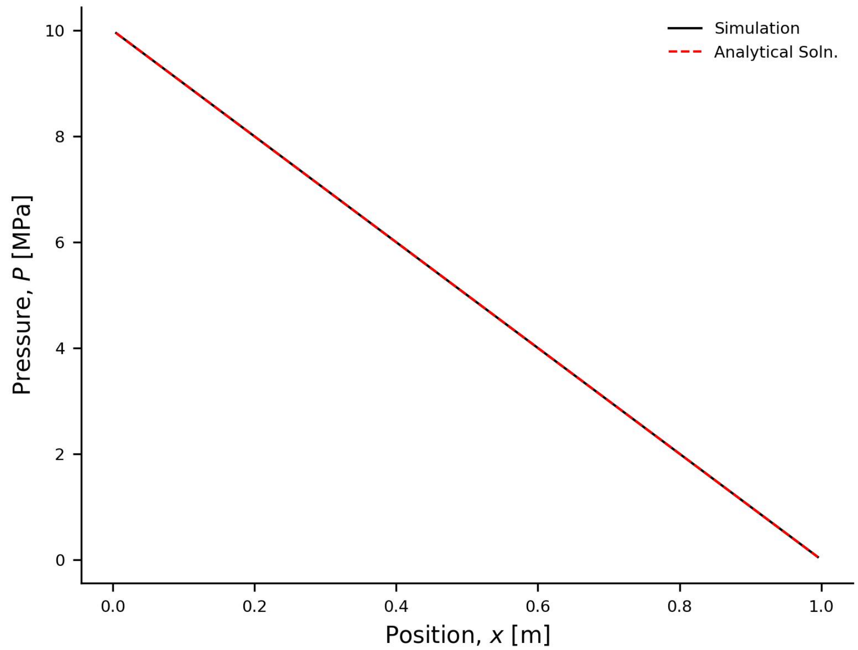

Figure 2: Comparison between analytical and numerical pressure solutions for the Python simulator.

The maximum flux relative error is defined as

$$E_q = \max_i \left| \frac{q_{\mathrm{num},i} - q_{\mathrm{ana},i}}{q_{\mathrm{ana},i}} \right|. \tag{6}$$

For the present simulation in MATLAB

$$E_P = 2.57 \times 10^{-14}, \quad E_q = 1.12 \times 10^{-13}. \tag{7}$$

In Python, the result is

$$E_P = 1.54 \times 10^{-14}, \quad E_q = 7.45 \times 10^{-14}. \tag{8}$$

7 Discussion

The numerical pressure distribution reproduces the analytical linear pressure solution and the numerical Darcy flux reproduces the analytical constant flux solution. The small residual errors arise from floating-point arithmetic and are negligible.

8 Conclusion

The simulator successfully reproduces the analytical solution for one-dimensional incompressible flow in a homogeneous porous medium. This provides verification that the finite volume implementation is correctly solving the governing equations for this benchmark problem.

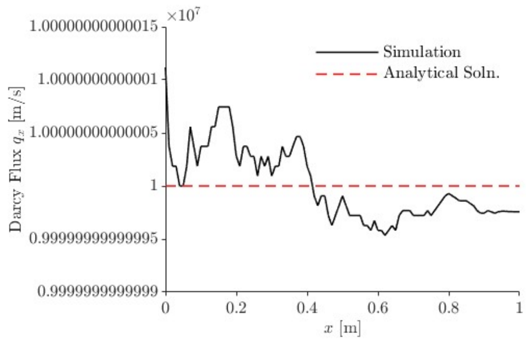

Figure 3: Comparison between analytical and numerical Darcy flux solutions for the MATLAB simulator.

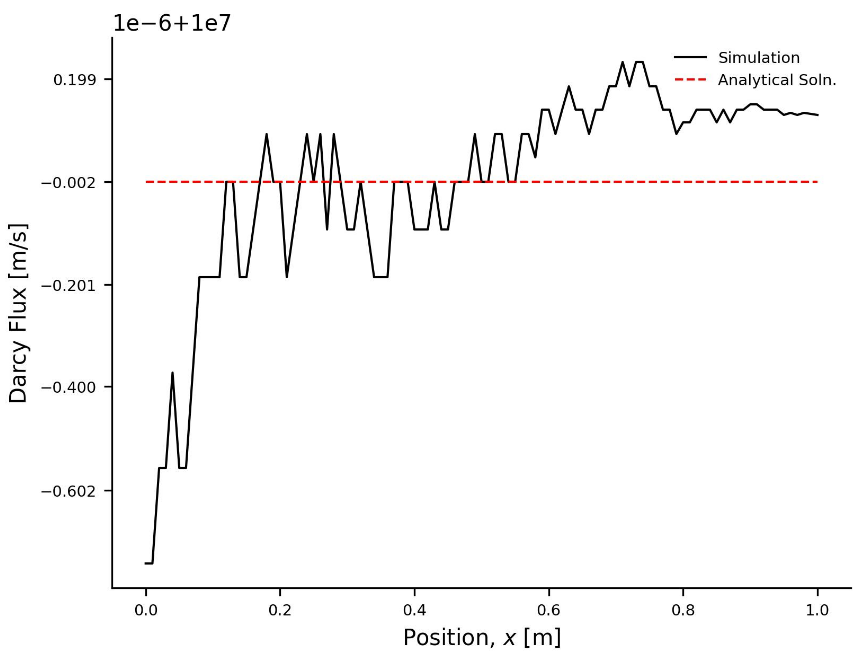

Figure 4: Comparison between analytical and numerical Darcy flux solutions for the Python simulator.